

NISHAL STANISLAUS SILVA, PhD

Computer Vision & ML Engineer | Industrial Inspection, IoT & Edge Deployment

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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PROFESSIONAL SUMMARY

Computer vision engineer with 5.5 years building and deploying production machine-vision systems on live manufacturing floors at MAS Holdings, South Asia's largest apparel manufacturer – training and evaluating vision models end to end, from raw camera footage through annotation, evaluation-framework design and deployment, on real production data rather than curated benchmarks. Automated a multilingual care-label QC system with OpenCV template and feature matching driving a PLC-controlled conveyor reject, cutting inspection time 99.5%, and a microscopic camera-array system for loom-side fabric defect detection that cut operator headcount 50%. A PhD in Information Engineering and Computer Science (University of Trento) added rigor in evaluation methodology, benchmark and ablation design, and real-time embedded inference: a published detector running at 14 ms on a Raspberry Pi 4, F1 0.76, 74x faster than a DTW baseline. Strong Python and PyTorch proficiency, hands-on OpenCV and ARM/embedded deployment experience under hard latency budgets, and a track record of investigating model failures against real-world footage rather than lab conditions. Canadian Permanent Resident, no sponsorship required, based in Toronto and open to relocating to Montreal.

CORE COMPETENCIES

Computer Vision & Perception: training and evaluating vision models on production video and camera footage (PyTorch, OpenCV), template and feature matching, image registration and warping, corner/edge detection, explainable defect overlays for human-in-the-loop QC, evaluation-framework design for real deployment conditions (precision/recall trade-offs, train/test protocol design).

Data & Model Lifecycle: owning training datasets end to end, from sourcing and annotation-quality review through failure-mode investigation on live footage; running end-to-end experiments; industrial and microscopic camera-array systems; multilingual label and print inspection.

Embedded & Real-Time Systems: Raspberry Pi and ARM CPU profiling, C++17 real-time inference engines, hard latency-budget deployment (14 ms on-device), PLC integration for automated reject/sort actuation, high-frequency IoT sensor ingestion and ETL, Linux/SSH-based development.

TECHNICAL STACK

Python, PyTorch, OpenCV, C++17, TensorFlow, Keras, scikit-learn, NumPy, pandas, Raspberry Pi / ARM, AWS, Git, Docker, CI/CD, pytest.

EDUCATION

PhD, Information Engineering and Computer Science – University of Trento, Italy	2021 - 2025
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK	2016 - 2018
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK	2013 - 2014

RESEARCH AND WORK EXPERIENCE

Independent Researcher | *Self-directed, Toronto, Canada*

Jan 2026 - Present

- Two peer-reviewed papers accepted (Smart Drums, JAES; MIRaaS, IEEE IS2 2026) covering real-time inference and edge deployment.

- Maintain Nebula, an open-source C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks; used AI-assisted development workflows, including Claude Code, throughout.

Postdoctoral Researcher | *University of Trento, Trento, Italy*

Jan 2025 - Dec 2025

- Owned the full ML pipeline for streaming audio and sensor data: acquisition, feature engineering, training and evaluation, edge deployment on Raspberry Pi 4, and monitoring.
- Published a real-time detector achieving 14 ms inference on Raspberry Pi 4, F1 0.76, 31.4% CPU utilization – 74x faster than a DTW baseline (JAES 2026).
- Designed frozen-protocol benchmark suites and ablation studies quantifying accuracy/latency/CPU trade-offs for embedded deployment.

Visiting Researcher | *McGill University, IDMIT/CIRMMT, Montreal, Canada*

May 2024 - Aug 2024

- Built a self-powered embedded sensor system (BNO055 IMU + HiFiBerry + Raspberry Pi 4); fused IMU and audio signals via a hierarchical finite-state machine, achieving F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka*

Jan 2024 - Mar 2024

- Conducted a human-centred evaluation of a real-time detection system against user footage and feedback (IEEE I3DA 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka*

Aug 2020 - Feb 2021

- Built ML and statistical pipelines for financial forensics, compliance monitoring, and anomaly detection on large structured enterprise data.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka*

Jan 2015 - Jul 2020

- Designed and deployed end-to-end computer-vision and IoT systems for manufacturing at scale, taking camera and sensor streams directly into production workflows; drove operational efficiency gains of up to 300%.
- Built an automated multilingual care-label QC system using OpenCV template and feature matching with an explainable defect overlay; iterated from handheld scanner to industrial camera to a conveyor with PLC-controlled reject, cutting inspection time 99.5%, with a human-in-the-loop path for borderline cases and continuous failure-mode review against production footage.
- Developed a loom-side fabric structural-defect detection system using a microscopic camera array, cutting operator headcount 50%.
- Owned training data end to end for these systems: sourcing, annotation review, and iterative retraining against real production footage rather than a fixed benchmark set.
- Introduced CI/CD and code review practices; mentored engineers and delivered group-wide computer-vision training.
- Designed and deployed a remote patient-vitals computer-vision monitoring device for COVID-19 hospital wards, recognized by Sri Lanka's Government Medical Officers' Association.

PUBLICATIONS

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3). Silva, N. & Turchet, L. (2026, in press). Smart Drums. *Journal of the Audio Engineering Society*.

LANGUAGES

English (native/C2) | Sinhala (mother tongue) | Italian (B1) | French (basic, A2, actively learning)